ANTARA THAPLIYAL

Hamilton, NJ | 640-225-3875 | thapliyalantara31@gmail.com | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Master's student in Computer Science at Rutgers University with 2 years of production software engineering experience. Skilled in Python, LangChain, LangGraph, MCP, prompt engineering, and GenAI evaluation, with hands-on experience building AI agents, agentic workflows, and RAG-based applications across vector stores, APIs, and LLM frameworks. Currently building multi-agent systems with real-time tool connectivity and evaluation pipelines; seeking a generative-AI engineering role.

TECHNICAL SKILLS

Languages & Databases: Python, SQL/MySQL, HTML, CSS
GenAI & Agent Frameworks: LangChain, LangGraph, Langfuse, MCP (Model Context Protocol), OpenAI API, CrewAI, FAISS, Chroma, RAG Architectures
ML Frameworks: PyTorch, TensorFlow, Pandas, NumPy
Cloud: AWS (S3, Bedrock, Bedrock AgentCore, CloudWatch)
Core Skills: AI Agent Development, Prompt Engineering & Robustness Testing, Agentic Workflow Design, GenAI Evaluation & Benchmarking, AI Guardrails & Safety, Explainability & Interpretability, Vector Stores/RAG, REST API Integration, Data Pipeline Design, Technical Documentation
Tools & Platforms: Git/GitHub, Azure DevOps, Jupyter Notebook, Streamlit, VS Code, PyCharm, Microsoft Office
Methodologies: Agile/Scrum, SDLC, Iterative Experimentation & Design

PROFESSIONAL EXPERIENCE

Otsuka Pharmaceutical, Princeton, NJ Jun 2026 – Aug 2026
GenAI Engineer Intern

- Built and shipped a **Code Review Agent** end-to-end using the internal skAI SDK pattern; fetches a target agent's codebase from Azure DevOps at runtime via REST API and reviews across three lenses: security (bandit), code quality (ruff), and skAI-compliance (Python AST), and lets each team supply its own review spec of custom checks; returns a risk-rated verdict, deployed to dev via the ADO CI/CD pipeline.
- Developed a **model-lifecycle agent** in Python that tracks AWS Bedrock (Claude) model EOL/legacy dates via boto3, applies tiered expiring-soon alerting, and emails managers on a scheduled run, surfaced through a Streamlit dashboard, reducing spend on deprecated or unused models against a per-user cost budget.
- Engineered a **multi-turn CLI** to invoke and interactively converse with agents on the platform, streamlining agent testing and iteration.
- Worked closely with the platform team on the internal **skAI SDK** and across the platform architecture: agents on **AWS Bedrock AgentCore** (registry, memory, gateway), agent code versioned in **Azure DevOps**, and **CloudWatch** monitoring.
- Executed manual **UAT** for the skAI platform as an Agent Consumer, identifying environment-readiness blockers and documenting them for the platform team to resolve.

Tata Consultancy Services (TCS), Pune, India Jul 2021 – Jun 2023
Software Engineer – Insurance Domain

- Partnered with cross-functional teams to design and deliver new product features for an enterprise platform, driving a 29% lift in sales.
- Engineered automated batch-processing data pipelines and letter-generation utilities, cutting manual effort by 30% and accelerating offline processing workflows.
- Built the "Autopay Re-collection" system, integrating database queries and REST API calls for automated loan repayment processing, improving collection efficiency.
- Maintained clear technical documentation, Git-based versioning workflows, and sprint documentation to ensure reproducibility and clarity across engineering teams.

PROJECTS

Shadow Board – Multi-Agent AI Decision System | Python, CrewAI, MCP, REST APIs, FastAPI, SSE | 2026

- Architected a multi-agent AI system with specialized personas (CFO, CMO, Legal, Devil's Advocate) that independently research via API-based tool integrations and engage in structured real-time agentic workflows with evaluation checkpoints; engineered persona-specific prompt strategies, robustness testing, and benchmarking harnesses using CrewAI.

Capital Finder Bot – RAG-Based AI Chatbot | LangChain, Langfuse, OpenAI API, FAISS, Streamlit, Python | 2025

- Built a RAG-powered chatbot with LangChain, OpenAI, and a FAISS vector store for context-aware retrieval; implemented RAGAS evaluation (faithfulness, answer relevancy) and LangFuse tracing of latency, token usage, and cost per query.

Abilify Clinical Q&A Agent – Agentic RAG System | Python, LangGraph, LangChain, LangFuse, RAGAS, OpenAI, Chroma | 2026

- Built a four-agent clinical Q&A system over Abilify's FDA label using LangGraph orchestration, with specialist agents (Clinical, Drug Interaction, Safety) using RAG retrieval (parent-child chunking, hybrid BM25 + vector search, Cohere reranking) and an inline LLM-as-judge agent that rechecks faithfulness, completeness, and citations and routes back for retry; LangFuse and RAGAS monitoring.

EDUCATION

Rutgers University, New Brunswick, NJ Expected Dec 2026
Master of Science in Computer Science | GPA: 4.0

G.B. Pant University of Agriculture & Technology, India Jun 2021
Bachelor of Technology in Information Technology

LEADERSHIP & ACTIVITIES

- **Certifications:** Supervised Machine Learning: Regression and Classification (Stanford Online & DeepLearning.AI, Coursera); Introduction to Model Context Protocol (Anthropic).
- **Astitva NGO (2017–2021):** Taught underprivileged children three days weekly.